



FORKLIFT OPERATIONS MANUAL

Safe handling and maintenance guide

Introduction

The **ESP32 Wi-Fi Controlled Forklift Robot** is a robotic model designed to demonstrate wireless movement and material-lifting operations. The robot is controlled using a smartphone or computer through a web-based interface hosted by the ESP32. It can move **forward, backward, left, and right**, along with raising and lowering the fork mechanism. The model demonstrates the practical application of **IoT, embedded systems, motor control, and robotics** in material-handling systems.

Key Features ESP32-based control system, Built-in Wi-Fi access point, Smartphone/web-browser control, Forward and backward movement, Left and right turning
Motorized forklift lifting mechanism, Real-time WebSocket communication, Dedicated emergency/stop controls, No external Wi-Fi router required, Simple and user-friendly control interface

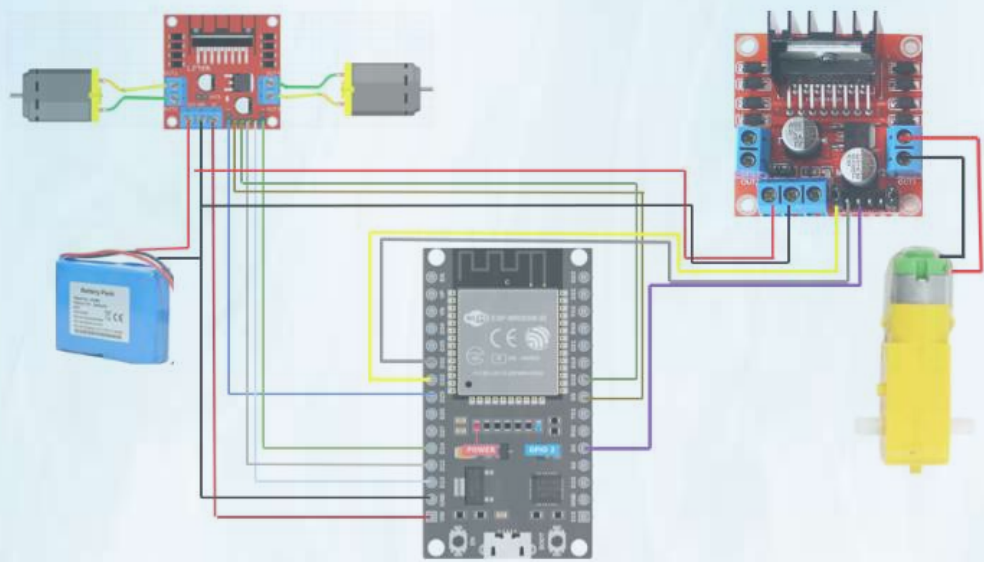
Objectives

The main objective of this kit is to provide students with hands-on learning in **robotics, ESP32 programming, Wi-Fi communication, and motor control**. It helps students understand how a forklift robot can be wirelessly controlled for **movement and lifting operations** while developing practical skills in **electronics, programming, and robotic system integration**.

Components

Sl. No.	Component	Quantity
1	ESP32 Development Board	1
2	BO Motor	5
3	L298N Motor Driver	2
4	Pin Header & Socket Headers	1 Set
5	Wheel	4
6	Connecting Wire	1/2 m
7	Jumper Wire (Male-Female)	1 Set
8	20 × 15 cm Double-Sided Dot-Hole PCB	1

Circuit diagram



Assembly Steps

Step 1: Required Components

3 BO motors, connecting wires, double-sided PCB, ESP32, soldering iron, solder wire, and PCB cutting tool.



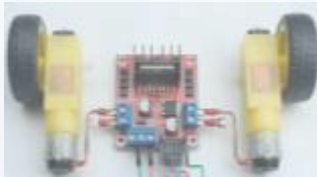
Step 2: Soldering

Solder wires to the 3 **BO motors** and secure the connections properly.



Step 3: Connect BO Motors to L298N Motor Driver

Connect the **BO motors to the L298N motor driver** by referring to the circuit diagram. Ensure the motor wires are connected to the correct output terminals and the connections are secure.



Step 4: Connect the remaining BO motor to the second L298N motor driver as per the circuit diagram.

ESP32 → L298N Connections

ESP32 Pin	L298N	Purpose
GPIO 25	ENA	Left motor speed
GPIO 5	IN1	Left motor direction
GPIO 18	IN2	Left motor direction
GPIO 14	ENB	Right motor speed
GPIO 12	IN3	Right motor direction
GPIO 13	IN4	Right motor direction
GPIO 33	ENA	Lift motor speed
GPIO 32	IN1	Lift motor direction
GPIO 4	IN2	Lift motor direction

For the second L298N (Lift System):

GPIO 33 → ENA
GPIO 32 → IN1
GPIO 4 → IN2

Step 5: Connect Battery

Connect the **battery positive (+) to the 12V terminal** of the L298N and the **battery negative (-) to GND**.

Step 6: Connect L298N to ESP32

Connect the **L298N 5V output to the ESP32 VIN pin** and **L298N GND to ESP32 GND**.

Step 7: Connect Control Pins

Connect all the **ESP32 control pins to the L298N motor drivers** according to the above connection table.

Step 8: Place the Circuit in the Case

Place the assembled circuit securely inside the **3D-printed case** and ensure all connections are properly arranged.

Step 9: Attach the Motors

Fix the **2 BO motors to the back side of the chassis** and secure them properly.

Step 10: Attach Motor Hubs

Connect the **two BO motor hubs to the front side of the chassis** and secure them properly.

Step 11: Attach Lift Motor

Connect the **remaining BO motor to the 3D-printed forklift case** as shown in the picture.



Step 12: Connect to Wi-Fi

Turn ON the forklift robot and connect your **mobile phone or laptop** to the Wi-Fi network using the following details:

Wi-Fi Name: Forklift_Robot_Pro

Password: 12345678

Step 13: Open Web Control

Open a **web browser** on your connected device and enter <http://192.168.4.1> in the address bar to access the forklift control interface.

Step 14: Control Panel

The **webpage and control panel** will open, allowing you to control the forklift robot **Forward, Backward, Left, Right, Stop, Lift Up, and Lift Down**.

Safety Precautions

- Check all wiring and connections before switching ON the power.
- Use the correct battery voltage for the circuit and motors.
- Keep hands and fingers away from the moving wheels and lift mechanism.
- Do not overload the forklift with heavy objects.
- Operate the robot only on a stable and flat surface.
- Stop the robot before adjusting or repairing any connections.
- Do not touch the circuit or motor terminals while the power is ON.
- Keep the robot away from water and moisture.
- Ensure the lifting mechanism does not exceed its mechanical limit.
- Switch OFF the battery when the robot is not in use.

Troubleshooting

- **Robot not moving:** Check the battery, motor-driver power supply, and motor connections.
- **One motor not working:** Check the BO motor wires and L298N output connections.
- **Robot moving in the wrong direction:** Reverse the motor wire connections.
- **Lift not moving:** Check the second L298N, lift motor, and lift control connections.
- **Lift moving in the wrong direction:** Swap the lift motor wires.
- **Wi-Fi not visible:** Check whether the ESP32 is powered ON and the program is uploaded correctly.
- **Webpage not opening:** Make sure the device is connected to Forklift_Robot_Pro and enter `http://192.168.4.1`.
- **Robot stops suddenly:** Check the battery voltage and power connections.
- **Control buttons not responding:** Refresh the webpage and reconnect to the robot's Wi-Fi.

Learning Outcomes

By using this kit, students will learn:

- Basics of **ESP32 programming and Wi-Fi communication**.
- How to control **DC motors using an L298N motor driver**.
- Basic concepts of **robotics, circuit connections, and motor control**.
- How to operate a robot through a **web-based control panel**.
- Practical skills in **assembling, testing, and troubleshooting** robotic systems.

Conclusion

The **ESP32 Wi-Fi Controlled Forklift Robot** provides students with a practical way to learn **robotics, programming, electronics, and wireless control**. The kit demonstrates real-time control of robot movement and lifting mechanisms through a web interface. It helps students develop hands-on skills in **circuit assembly, motor control, programming, testing, and troubleshooting**. The model also encourages students to understand how robotics can be applied to **real-world material-handling applications**. It provides a strong foundation for students to develop more advanced **IoT and autonomous robotic projects**.